

Course: SIT222 - Spring 2025-2026 Assignment: Network Topology & Infrastructure Design

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1. Introduction and Design Assumptions

This report outlines the proposed network infrastructure design for the new 3-floor Faculty of Computing & AI academic building. The design strictly adheres to TIA-568 and TIA-569 standards, guaranteeing 100% wired connectivity for all current devices while incorporating a mandatory 30% spare capacity for future expansion.

2. Physical Infrastructure and Pathway Routing (TIA-569)

The building's reinforced concrete construction presents significant routing constraints. To avoid compromising the structural integrity, horizontal cabling will be routed using suspended cable trays, taking advantage of the 3.5-meter ceiling height. To reach wall-mounted faceplates, cabling will drop down from the ceiling trays through surface-mounted PVC conduits and wall ducts.



Figure A: Low-profile surface-mounted PVC (Permanent Virtual Circuit) trunking terminating into network faceplates (white squares). This narrow profile will be utilized for individual staff offices to provide clean, unobtrusive drops to user workstations.



Figure B: High-capacity surface-mounted PVC trunking. Wider profiles will be utilized in high-density areas (such as the 40-PC Computer Labs) and along primary corridors to accommodate bulk cable runs and safely support the mandatory 30% future load growth capacity.

Given the substantial floor area of approximately 1,500 m² per floor, a dedicated Telecommunications Room (TR) will be established near the physical center of each floor. This central placement guarantees that no horizontal cable run exceeds the standard 90-meter maximum length.



Figure C: Proposed layout for the floor-level Telecommunications Rooms (TR). This TIA-568 compliant design demonstrates proper horizontal wire management, permanently terminating structural wall cabling at the top patch panels and utilizing short, flexible patch cords to connect to the active access switches below.

3. Equipment and Cabling Specifications

To meet both current demands and the expected 15-20 year lifespan of the infrastructure, the following materials have been selected:

- **Horizontal Cabling:** Cat6a Unshielded Twisted Pair (UTP) will be deployed to support the mandatory 1 Gbps to the desktop and ensure future support for 10 Gbps endpoint connections.

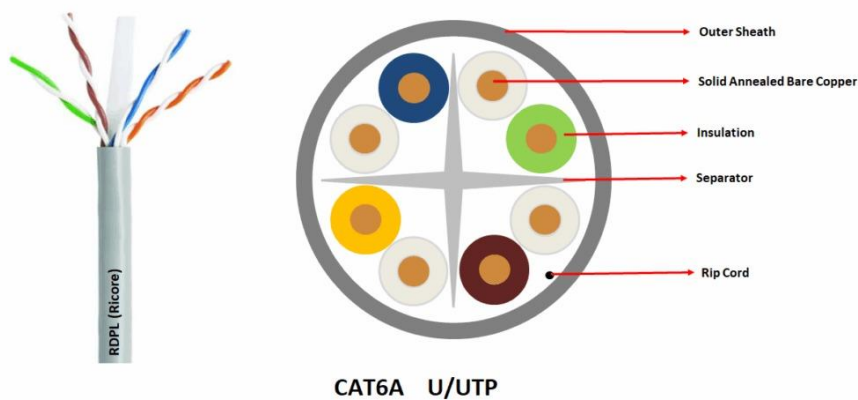


Figure D: Cat6a Unshielded Twisted Pair (U/UTP) cable cross-section. The internal plastic separator (spline) physically isolates the copper pairs to eliminate crosstalk, ensuring reliable 10 Gbps bandwidth capacity to the desktop.

- **Backbone Cabling:** OM4 Multimode Fiber Optic cable will be used for vertical routing to satisfy the 10 Gbps backbone requirement.



Figure E: OM4 Multimode Fiber Optic patch cord (Erika Violet) with LC (Lucent Connector) duplex connectors. The Erika Violet jacket is utilized to visually distinguish the 10 Gbps OM4 backbone runs from legacy OM3 infrastructure, ensuring zero confusion within the Telecommunications Rooms and main Server Room.

- **Patch Panels:** All incoming Cat6a horizontal cables will terminate into 48-port patch panels located in the TRs. Short, stranded patch cords will then connect the panels to the switches. This prevents physical strain on the switch ports and establishes a rigid cable labeling system.

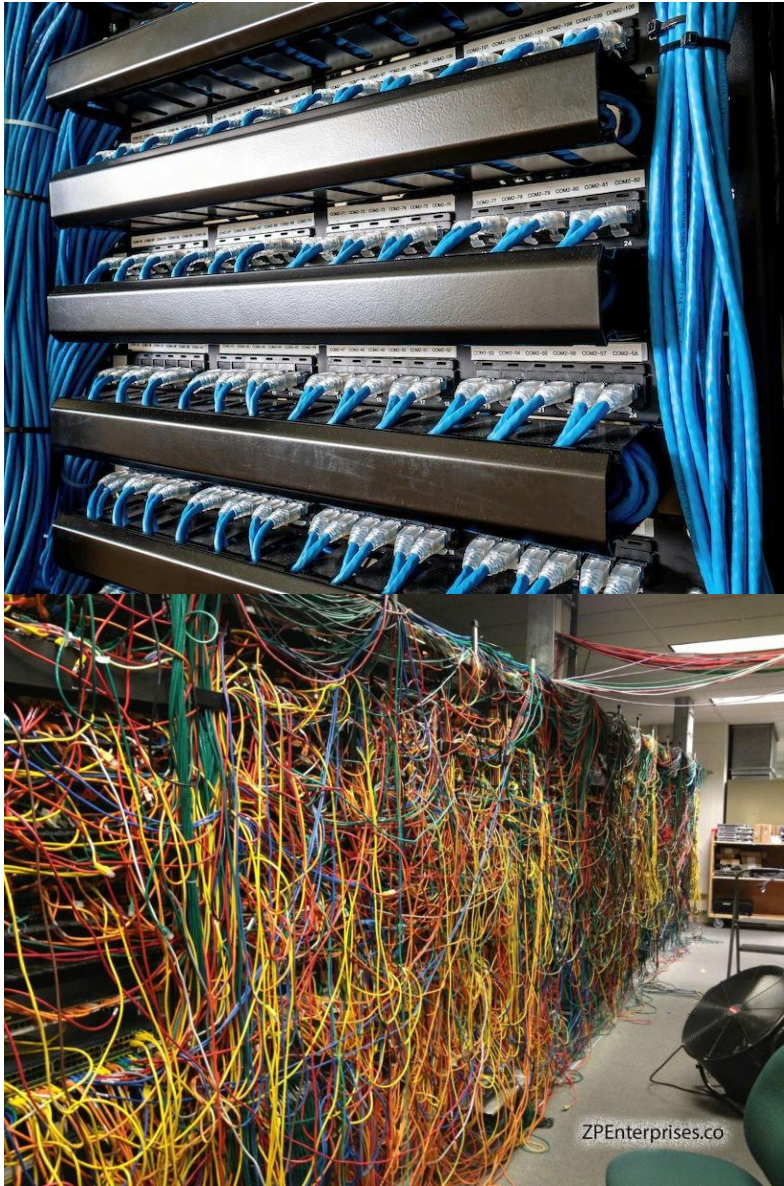


Figure F (Proposed TIA-568 Compliant Design): Terminating horizontal cables at a patch panel. Our design dictates that all stiff, solid-core wall cables permanently terminate into the rear of a 48-port patch panel. We then use short, flexible, stranded-copper 'patch cords' to connect the front of the panel to the access switch. This completely removes the physical strain from the expensive switch ports. Crucially, it allows for permanent, fixed cable labeling, and makes Moves, Adds, and Changes (MACs) effortless, if port needs to be reassigned, we simply move a small flexible patch cord rather than pulling a new cable through the building.

Figure G (Incorrect Design): Connecting horizontal wall cables directly into a switch. In building infrastructure, cables running through ceilings are solid-core copper. They are heavy, stiff, and inflexible. Plugging them directly into a switch causes mechanical stress. The weight of the cables will eventually bend the pins and break the fragile ports on an expensive switch. Furthermore, if a user moves offices, the entire stiff cable must be re-routed, leading to a disorganized 'spaghetti' of cables that makes troubleshooting impossible.

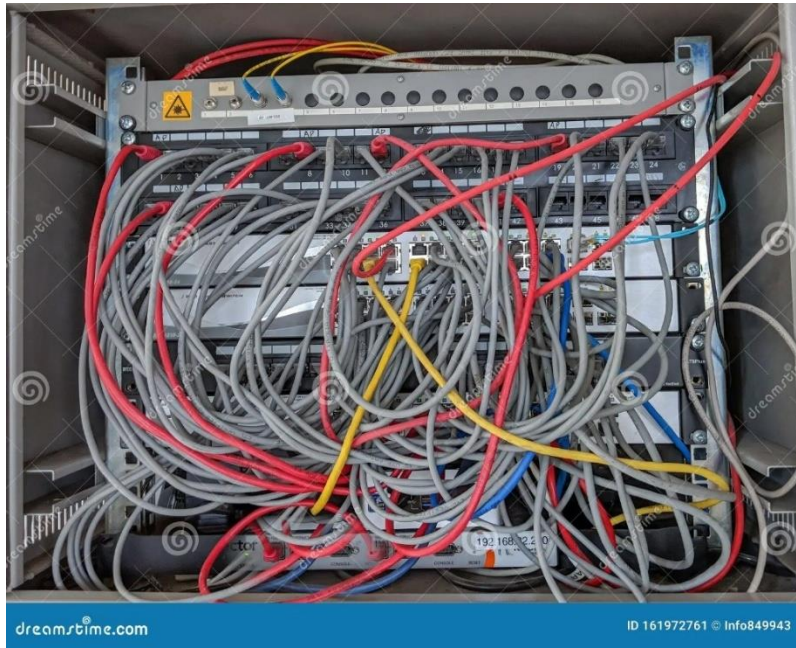


Figure H (Small-Scale Incorrect Design): Even in smaller, localized distribution boxes, bypassing a patch panel leads to immediate port blocking and hazardous cable strain on the active hardware.

- **Hardware Selection:** * Access Layer: 48-port Layer 2 PoE+ Gigabit switches.

Figure I: 48-Port PoE+ Access Switch. Provides 1 Gbps access to individual workstations, Power over Ethernet (PoE+) for IP phones and wireless access points, and features dedicated 10G SFP+ uplink slots for high-speed fiber connectivity to the core network.



- Distribution & Core Layers: Layer 3 routing switches located in the ground floor server room.
- Transceivers: 10GBASE-SR SFP+ modules will be utilized to convert and connect the fiber optic backbone into the copper-based switches.

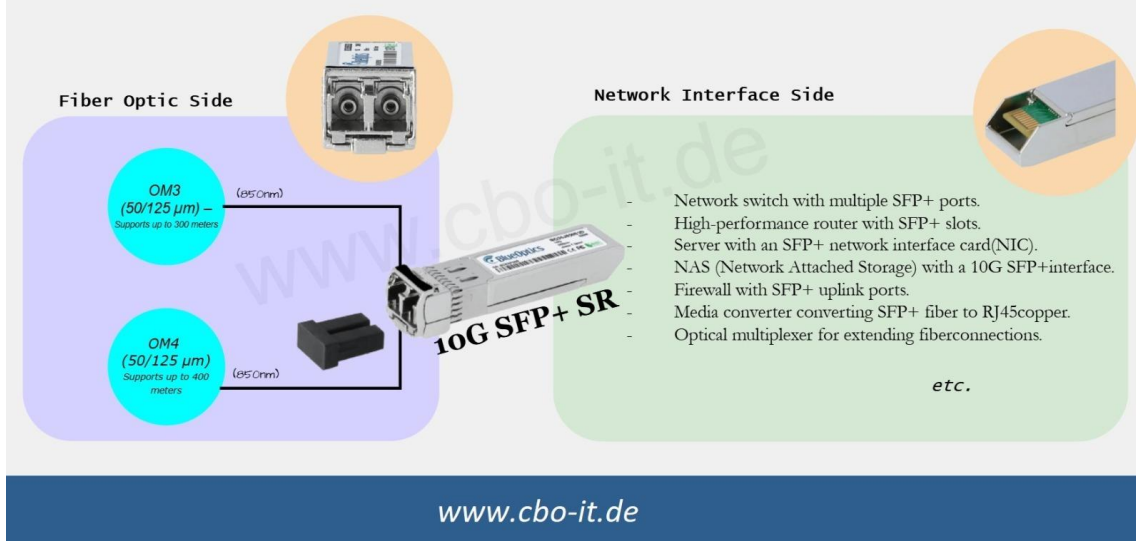


Figure J: 10GBASE-SR SFP+ Transceiver. This module is utilized to convert the OM4 optical fiber backbone connection to interface with the active switch uplink ports, enabling high-speed vertical routing between the floor-level TRs and the Ground Floor Server Room.

4. Outlet and Port Calculations

Based on TIA-568 recommendations, the baseline metric for calculations is a minimum of 2 outlets per work area or one outlet per 10 m². To ensure accurate hardware sizing, a 48-port switch is calculated as having 44 usable endpoint ports, reserving 4 ports for redundant fiber uplinks.

Calculation Methodology per Room Type:

- **Lecture Halls/Tutorials:** 4 ports (Instructor PC, IP Phone, AV/Auxiliary).
- **Computer Labs:** 1 port per student PC, 2 instructor ports, 2 printer ports.
- **Offices:** 3 ports (Workstation, IP Phone, Printer/Auxiliary).
- **Future Growth:** A strict 1.30 multiplier (30%) is applied to the base point totals to meet the required spare capacity.

4.1 Ground Floor

- Rooms: 2 Lecture Halls (100 seats), Reception/Admin (5 staff).
- Base Points: 20 ports.

- With 30% Spare Capacity: 26 ports required.
- Hardware Required: **1x 48-Port Access Switch.**

4.2 First Floor

- Rooms: 1 Lecture Hall (80 seats), 4 Tutorial Rooms, 2 Computer Labs (40 PCs), 10 Offices, Server Room, Meeting Room .
- Base Points: 146 ports.
- With 30% Spare Capacity: 190 ports required.
- Hardware Required: **5x 48-Port Access Switches.**

4.3 Second Floor

- Rooms: 1 Lecture Hall (80 seats), 4 Tutorial Rooms, 2 Computer Labs (40 PCs), 15 Offices, Research Lab .
- Base Points: 173 ports.
- With 30% Spare Capacity: 225 ports required.
- Hardware Required: **6x 48-Port Access Switches.**

4.4 Third Floor

- Rooms: 6 Tutorial Rooms, 3 Computer Labs (30 PCs), 20 Offices, Library/Study Area .
- Base Points: 216 ports.
- With 30% Spare Capacity: 281 ports required.
- Hardware Required: **7x 48-Port Access Switches.**

Total Access Hardware: 19x 48-Port Access Switches

5. Network Topology & Redundant Architecture

To ensure high availability and efficient traffic management across all three floors, our team mapped out the logical framework of the building. We began by drafting a precise structural map of the network's active layers to calculate optimal cable paths and switch stacking requirements.

5.1 Initial Structural Design Blueprint (Developed by Team)

Below is our hand-built structural diagram mapping the physical distribution of hardware across the Ground Floor Server Room and the Telecommunications Rooms (TR) on each individual floor.

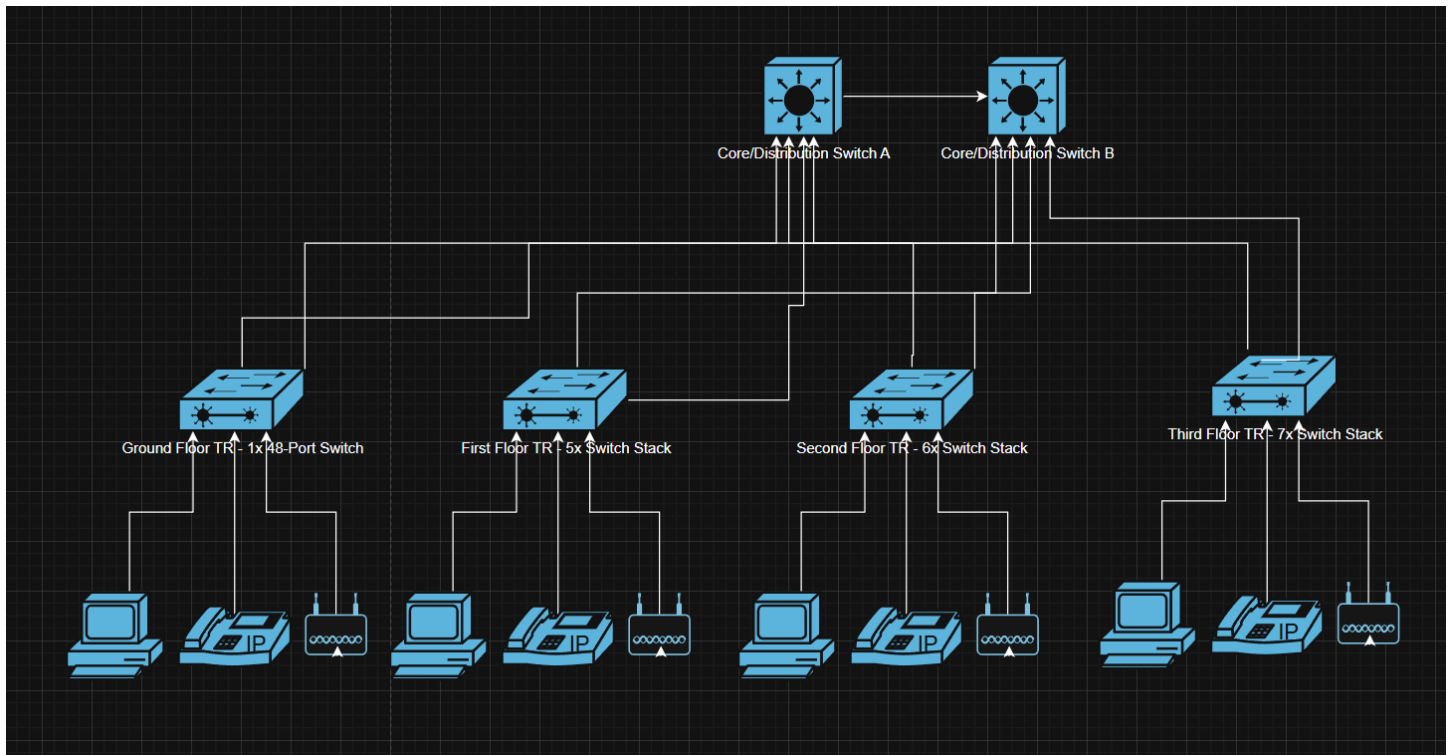


Figure K1: Team-Developed Physical Distribution Blueprint. This initial drafting shows the physical placement of the 19 access switches grouped into logical floor-level stacks, along with the designated Cat6a UTP endpoint connections to workstations, IP phones, and WAPs.

Draw.io Link:

<https://drive.google.com/file/d/1TZiWWHLNKT90Ewq61RZbJY92EjyOBb7Y/view?usp=sharing>

5.2 Final High-Fidelity Enterprise Topology Diagram

Building upon our initial layout, we translated the physical metrics and standard TIA-568/569 requirements into a comprehensive enterprise network topology. This model highlights the redundant links and specific backbone media that will be deployed in the building.

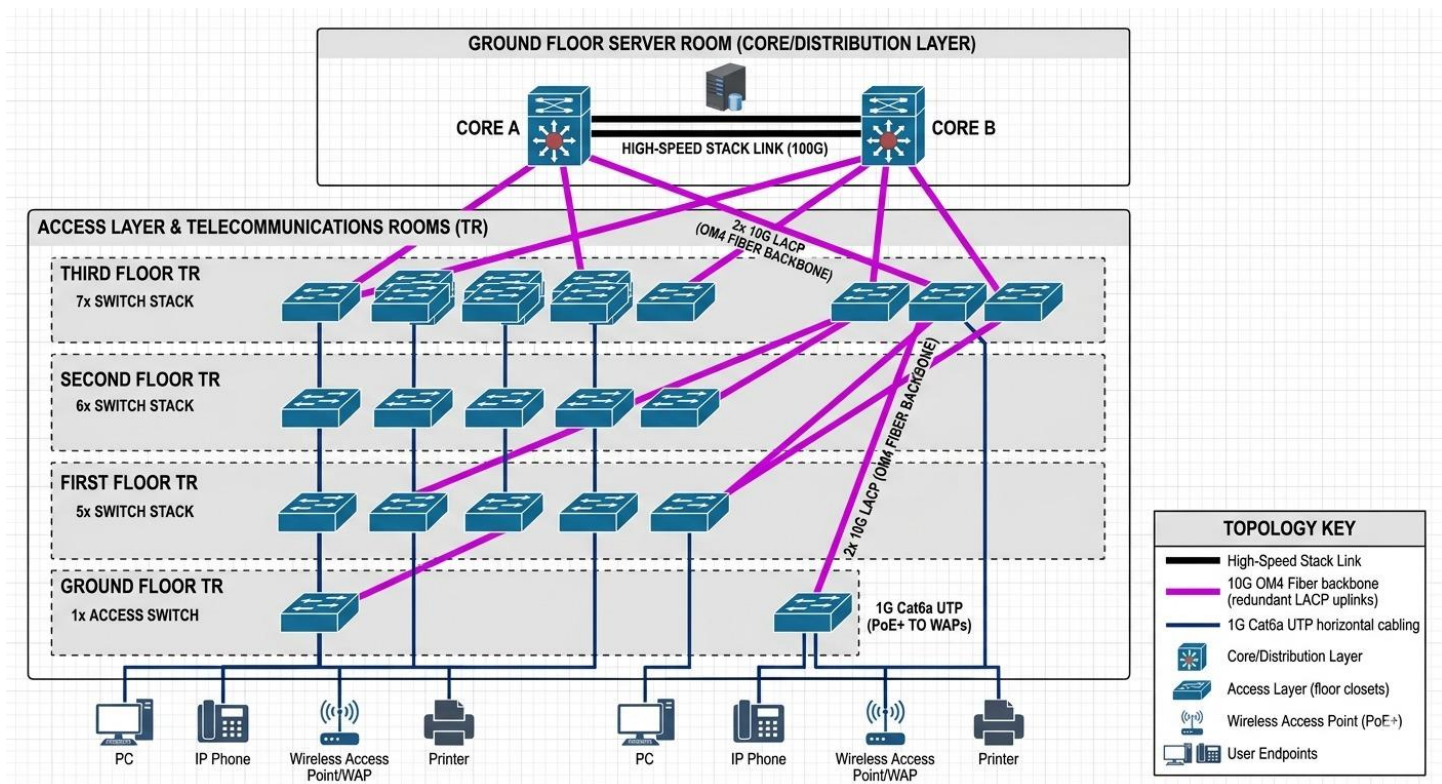


Figure K2: Final Three-Tier Hierarchical Network Topology. Our completed design illustrates the high-speed 100 Gbps Core Switch stack link, the active/active dual 10 Gbps OM4 Erika Violet fiber-optic uplinks per floor, and the redundant Link Aggregation (LACP) failover system.

6. Supplemental Wireless Network Design (Future-Proofing & Mobility)

While the core infrastructure guarantees 100% wired connectivity to fixed desktop endpoints, modern educational environments demand seamless, high-speed mobility for student laptops, tablets, and mobile devices.

To expand on the core TIA-569 infrastructure, a complementary wireless layer has been designed utilizing Enterprise Wi-Fi 6E/7 Access Points (APs).

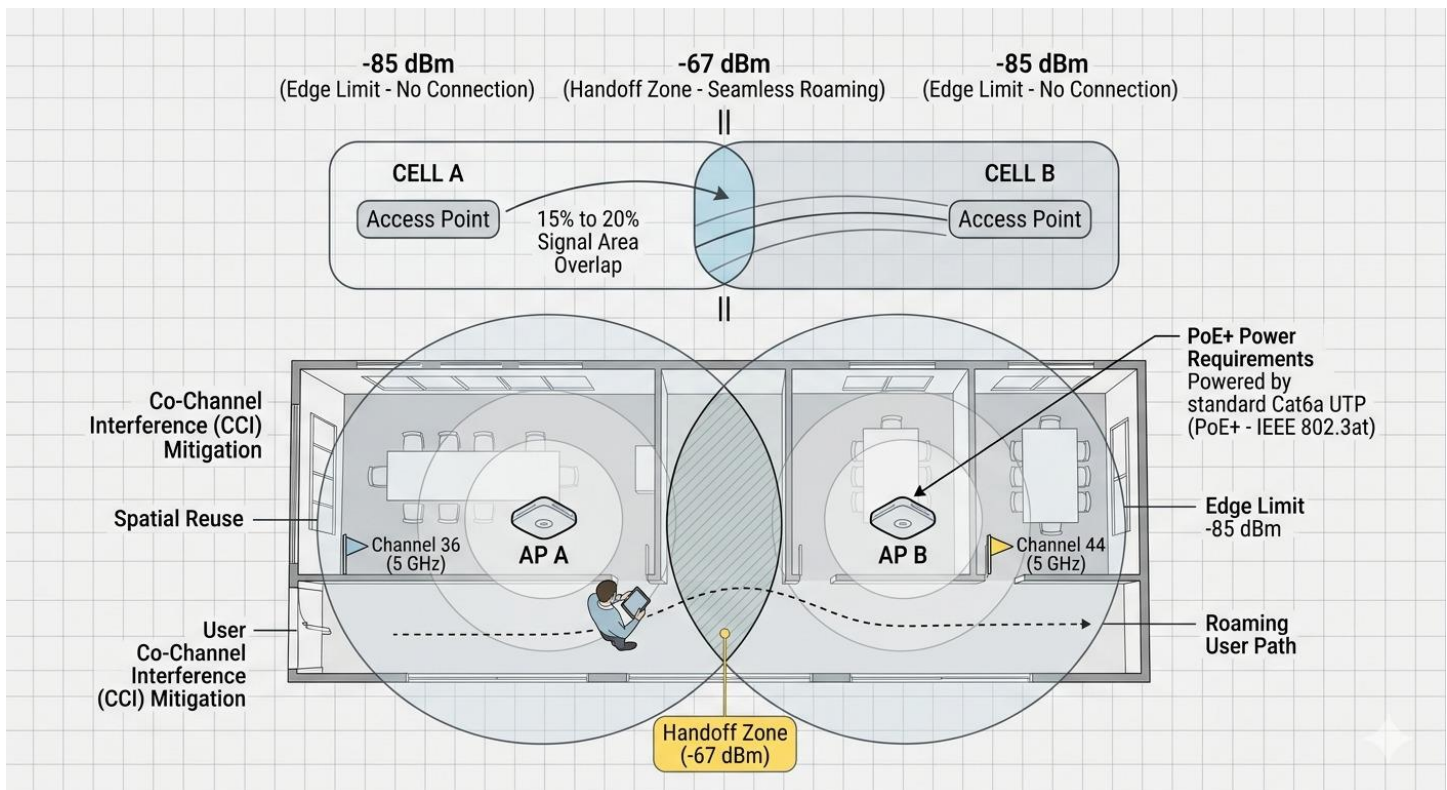


Figure L: Enterprise Wireless Cell Deployment and Roaming Mechanics. The diagram illustrates how careful physical Access Point placement ensures a specific -67 dBm signal handoff zone. The floorplan example shows how spatial reuse (alternating Channels 36 and 44) mitigates Co-Channel Interference while guaranteeing a seamless connection for the roaming user.

6.1 Spatial Reuse and the 15–20% Cellular Overlap Principle

To prevent dropped connections, high latency, or session timeouts when students move between lecture halls and tutorial rooms, the physical placement of the ceiling-mounted Access Points follows the strict **15% to 20% cellular overlap rule**.

- **The Problem of Cell Edges:** As a client device moves away from Access Point A, its received signal strength indicator (RSSI) drops. Without a calculated overlap, the device will cling to a weak AP (known as a "sticky client") until the signal is completely dropped before looking for a new AP.
- **The Solution (Calculated Overlap):** Access Points are physically positioned so that their coverage circles overlap by **15–20%**. When the client enters this specific overlap region, the RSSI drops to approximately **-67 dBm**. At this precise threshold, the client's internal wireless card automatically initiates a

sub-second roaming handoff to Access Point B, ensuring uninterrupted connectivity during continuous video streams or active virtual lab sessions.

6.2 Frequency Selection and Co-Channel Interference (CCI) Mitigation

Overlapping physical coverage creates a secondary engineering challenge: **Co-Channel Interference (CCI)**. To prevent two neighboring APs from drowning each other out, our design uses a strict channel reuse scheme:

- **Non-Overlapping Channels:** On the 2.4GHz band, APs are strictly assigned to channels **1, 6, and 11**.
- **Dynamic Frequency Selection on 5 GHz & 6 GHz:** Neighboring access points within the same floor corridor are mapped to alternate, non-overlapping channels (e.g., Channels 36, 44, 52).
- **PoE+ Power Requirements:** To simplify physical deployment, all ceiling-mounted APs will run over standard **Cat6a UTP horizontal cabling**, drawing power directly from the 48-port Layer 2 Access switches via **Power over Ethernet (PoE+ - IEEE 802.3at)**.